Heuristic Ramp-Metering Coordination Strategy Implemented at Monash Freeway, Australia

Ioannis Papamichail, Markos Papageorgiou, Vincent Vong, and John Gaffney

A new traffic-responsive feedback control strategy, heuristic ramp-metering coordination (HERO) that coordinates local ramp-metering actions in freeway networks is presented. The proposed coordination scheme is simple and reactive [i.e., based on readily available real-time measurements without the need for real-time model calculations or external disturbance (e.g., demand) prediction]. HERO employs an extended version of the feedback regulator ALINEA at the local level. The new strategy overcomes the problem of uncertain freeway capacity by targeting the critical occupancy for maximum throughput. HERO outperforms uncoordinated local ramp metering and approaches the efficiency of sophisticated optimal control schemes. HERO has been implemented by VicRoads, the road authority for the state of Victoria, Australia, at six consecutive inbound on-ramps on the Monash Freeway in Melbourne, Australia. This pilot project is part of the Monash–CityLink–West Gate (MCW) upgrade. The obtained results show a significant increase of traffic throughput and a reduction of travel times compared with the previously used ramp-metering system. To maximize performance across the entire 75-km route of the MCW upgrade, HERO is currently under implementation at 63 on-ramps.

Ramp metering aims at improving traffic conditions by appropriately regulating inflow from on-ramps to the freeway mainstream. Traffic-responsive ramp-metering strategies, as opposed to fixed-time strategies, are based on real-time measurements from sensors installed in the freeway network and can be classified as local or coordinated.

Local ramp-metering strategies make use of measurements from the vicinity of a single ramp to control each on-ramp independently of all other metered ramps in the network. Most prominent examples of local ramp-metering strategies are the demand–capacity (DC) and the occupancy (OCC) strategies (*1*) and the ALINEA strategy (*2, 3*) and its variations (*4, 5*). DC and OCC are feedforward disturbance–rejection schemes that target explicitly the flow capacity in the merge area on the basis of mainstream measurements of flow or occupancy, respectively, upstream of the ramp. Recent works (*6–8*) have demonstrated that the real flow capacity in a merge area may vary quite substantially from day to day even under similar environmental conditions. Naturally, highway capacity differences become even more pronounced in the case of adverse environmental conditions (*9, 10*). Thus, any ramp-metering strategy attempting to achieve a prespecified capacity flow value may lead to either overload and congestion (on days when the real capacity happens to be lower than its prespecified target value) or to underutilization of the infrastructure (on days when the real capacity happens to be higher than its prespecified target value).

In contrast, critical occupancy, at which capacity flow occurs, seems to be more stable (*8*), even under adverse weather conditions (*9, 10*). The ALINEA strategy and its variations are feedback control schemes targeting a set point (typically the critical value) for downstream occupancy. It is probably because of this feedback structure that targets critical occupancy that ALINEA was found to lead to significantly better performance than DC and OCC strategies in several comparative field evaluations (*3*).

Coordinated ramp-metering strategies make use of measurements from an entire freeway network to control all metered ramps within the network. Coordinated strategies may be more efficient than local ramp-metering strategies when multiple bottlenecks on the freeway or restricted ramp storage spaces exist. Coordinated ramp-metering approaches include multivariable control strategies (*11, 12*) and optimal control strategies (*13–21*). Multivariable regulators are derived from linearization of the strongly nonlinear traffic flow models, which limits their efficiency in case of heavy congestion. However, optimal control approaches employ relatively complex numerical solution algorithms that may be a burden for field application. This may be the main reason that field-implemented coordinated ramp-metering strategies up to now have been based on heuristic rule-based approaches.

Bogenberger and May (*22*) presented an extensive review of heuristic coordinated traffic-responsive ramp-metering algorithms. A number of studies have compared the performances of these algorithms. Recently, these comparisons were summarized (*23*) for a number of heuristic coordinated strategies that have been implemented in United States, including the zone and the stratified zone algorithms, the bottleneck algorithm and the helper algorithm. The main drawbacks of these approaches are as follows:

- Most algorithms employ a feedforward (rather than feedback) approach at either the local level, the coordination level, or both; this approach may lead to increased sensitivity with respect to various unexpected disturbances.
- All algorithms target prespecified flow capacity values that, in view of the real capacity variations from day to day, may lead to either overload or underutilization of the freeway infrastructure.

In view of these drawbacks, it would be desirable to have a coordinated ramp-metering strategy that possesses the following features:

I. Papamichail and M. Papageorgiou, Dynamic Systems and Simulation Laboratory, Department of Production Engineering and Management, Technical University of Crete, 73100 Chania, Greece. V. Vong and J. Gaffney, Monash–CityLink–West Gate Upgrade, VicRoads, 3 Prospect Hill Road, Camberwell, Victoria 3124, Australia. Corresponding author: M. Papageorgiou, markos@dssl.tuc.gr.

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- It should coordinate local ramp-metering actions in a suitable way so as to avoid the drawbacks of uncoordinated application.
- Involved algorithms should be feedback based to reduce sensitivity to unexpected disturbances.
- It should be simple and transparent (e.g., rule based).
- It should be reactive, so that no real-time model involvement and no external disturbance (e.g., demand) prediction are needed.
- It should approach the efficiency of sophisticated optimal control schemes.
- It should be generic (i.e., directly applicable to any freeway network) without a need for cumbersome parameter calibration or fine-tuning.

A new heuristic feedback control strategy that is responsive to traffic and coordinates local ramp-metering actions in freeway networks was indeed developed in an attempt to fulfill the requirements mentioned above. The strategy was named heuristic ramp-metering coordination (HERO) and it was extensively tested by simulation (*24, 25*) as well as in field implementations. The proposed coordination scheme is simple and utterly reactive (i.e., based on readily available real-time measurements), without the need for real-time model calculations or external disturbance prediction. HERO employs an extended version of the feedback regulator ALINEA at a local level. The new strategy targets the critical occupancy for throughput maximization that is deemed more robust than targeting a prespecified capacity value. HERO was found (*24, 25*) to outperform uncoordinated local ramp metering and to approach the efficiency of sophisticated optimal control schemes without the need for external disturbance prediction.

The rest of this paper is organized as follows. The modular structure of the generic HERO coordination strategy and software is presented first. This is followed by a discussion of the field application of HERO at the Monash Freeway in Melbourne, Australia, while some field evaluation results are presented next. The paper ends with a summary of the main conclusions.

HERO COORDINATION STRATEGY AND SOFTWARE

A generic software has been developed that implements the HERO coordination scheme for any freeway network by suitable input configuration. The modular structure of this software is presented in Figure 1. As the figure indicates, one ALINEA block is assigned to each metered on-ramp; this constitutes the (functionally) local part of the overall strategy that comprises several modules (outlined below) with corresponding specific tasks. The local ALINEA blocks are coordinated by the HERO scheme, as indicated in Figure 1. The overall control software may be implemented in a central control room (as in the Monash Freeway application) with dual communication lines to receive the real-time measurements and to submit the control decisions, respectively. Alternatively, the ALINEA blocks may be installed locally at each ramp while the HERO module is installed centrally.

Figure 1 also indicates the measurement requirements for HERO operation, which comprise three measurement locations at each metered on-ramp (at both on-ramp boundaries as well as around the

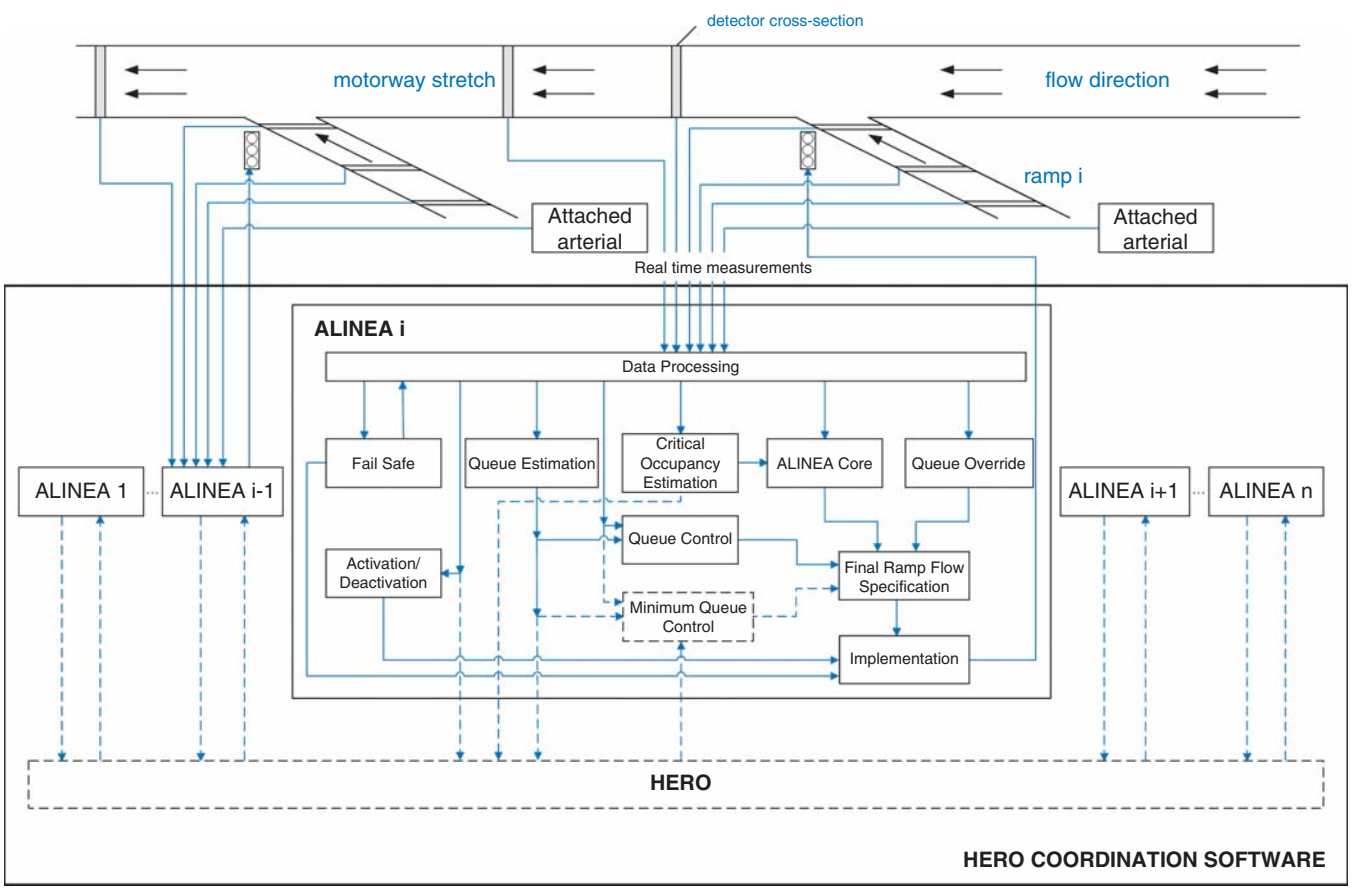


FIGURE 1 Modular structure of HERO coordination software.

middle of the on-ramp) and one mainstream measurement, preferably downstream of the on-ramp nose (as required by ALINEA). In case of a mainstream bottleneck downstream of the on-ramp merge area with a capacity lower than the merge area capacity, an additional mainstream measurement from the downstream bottleneck location may be required. The particular modules are outlined in what follows.

Real-time data from the mainstream and the on-ramps are appropriately averaged and smoothed by the data processing module, while the fail safe module decides on possible graceful-degradation actions in case of measurement failures. The activation–deactivation module switches the signal control on or off according to respective preset traffic conditions at the mainline.

The ALINEA core module calculates the desired ramp exit flow at each ramp for local maximization of the mainstream throughput, according to the ALINEA strategy. ALINEA is an I-type feedback control scheme targeting a set point (typically the critical value) for the downstream occupancy. ALINEA with set point $\tilde{o}$ reads

$$q_{\mathrm{AL}}(k) = q_{\mathrm{AL}}(k-1) + K_I\left[\tilde{o} - o(k-1)\right] \tag{1}$$

where

$k = 1, 2, \ldots, n$ is discrete time index of corresponding control period T of 15, . . . , 60 s,
$o(k-1)$ = merge area occupancy averaged over the last control period T,
$K_I > 0$ = I-feedback gain factor, and
$q_{\mathrm{AL}}(k)$ = ALINEA ramp exit flow ordered by the regulator for the next control period.

The calculated $q_{\mathrm{AL}}(k)$ is eventually bounded; the lower bound is given by the minimum admissible ramp flow $q_{\min}$, while the upper bound is given by the ramp's flow capacity (under signal control) $q_{\max}$. To avoid the well-known wind-up phenomenon of I-type regulators, (i.e., continued accumulation of the regulation error when control constraints are active), the term $q_{\mathrm{AL}}(k-1)$ in Equation 1 is also bounded accordingly. Related field results are presented in the following section.

Since its development in the late 1980s, ALINEA has been successfully implemented at hundreds of ramps around the world. ALINEA has been extended recently and is now accompanied by a suite of algorithms (see Figure 1) with some innovative features outlined in the corresponding module descriptions below.

In case measurements from the merge area are not available, upstream measurements are used by a variation of ALINEA called UP-ALINEA (*4*). In case of a bottleneck situated downstream of the merge area and featuring a capacity lower than the merge area capacity, ALINEA must be fed with measurements collected at the downstream bottleneck. The control oscillations that may be observed in this case are avoided by using ALINEA's proportional-integral extension (*26*). This kind of bottleneck may be due to a lane drop, a strong slope, a curvature, a speed limit, a strong weaving zone, or an uncontrolled downstream ramp. Finally, in case the targeted critical occupancy is not known, it can be automatically estimated (and updated) by activating the critical-occupancy estimation module, which involves a Kalman filter–based estimator (*5*).

The application of any ramp-metering strategy may lead to the creation of ramp queues. If a queue covers the whole on-ramp space and queue spillover is imminent, appropriate queue management actions are necessary to avoid interference with the adjacent street network. One of the queue management strategies presented below needs an estimated queue length on the on-ramp. The ramp queue length estimates can be calculated by using flow measurements from the entrance and the exit of the ramp and occupancy measurements from the middle of the ramp. Occupancy measurements are used to add a correction term to the estimation that would otherwise be purely based on flow measurement balance, thus accumulating any flow measurement errors. This queue estimation procedure is based on a Kalman filter approach [see Vigos et al. (*27*) for details] and is included in the queue estimation module. The estimated number of vehicles (the queue) $\hat{N}(k)$ is given by

$$\hat{N}(k) = \hat{N}(k-1) + T\left[q_{\mathrm{in}}(k-1) - q_{\mathrm{out}}(k-1)\right] + K_{\mathrm{est}}\left[N^m(k-1) - \hat{N}(k-1)\right] \tag{2}$$

where

$q_{\mathrm{in}}(k-1)$, $q_{\mathrm{out}}(k-1)$ = flow measurements at entrance and exit, respectively, of ramp over last control period T,
K_{est} = gain parameter of the filter, and

$$N^m(k-1) = N^{\mathrm{BB}}\frac{L_{\mathrm{ph}}}{L_{\mathrm{ph}} + \epsilon_{\mathrm{el}}}o_{\mathrm{mid}}(k-1)$$

where

N^{BB} = bumper-to-bumper capacity of ramp,
L_{ph} = average physical length of vehicles,
$o_{\mathrm{mid}}(k-1)$ = occupancy measurement (lane average) at middle of ramp over the last control period T, and
ϵ_{el} = electrical length of detectors placed at middle of ramp.

The produced estimates are eventually bounded; the lower bound is zero while the upper bound is the maximum number of vehicles N^{cap} that can be accommodated in the ramp at standstill, including the usual safety distance between vehicles (e.g., 1 m).

Remarkably, this ramp-queue estimation method was taken over by other researchers who tested it in the field and compared it with two alternative estimation schemes (*28*); the reported results showed a clear superiority of the method versus both alternative schemes.

Creation of long ramp queues that would interfere with adjacent street traffic is avoided with the application of an ordinary queue override or a queue control (*4*) policy, whose decisions may supersede those of ALINEA if the respective calculated queue override ramp exit flow or queue control ramp exit flow are higher than $q_{\mathrm{AL}}(k)$. Both queue management policies are implemented in the queue override module and the queue control module, respectively. Queue override is decided from occupancy measurements at the entrance of the on-ramp. If that occupancy exceeds a threshold, o_{th}, the queue override ramp exit flow, $q_{\mathrm{QO}}(k)$, is set to a (prespecified) high flow value, q_{over}, to reduce the queue; otherwise, the queue override exit flow is set to zero. In contrast, the queue control policy applies the following deadbeat controller [see Smaragdis and Papageorgiou (*4*) for details]:

$$q_{\mathrm{QC}}(k) = -\frac{1}{T}\left[N_{\max} - \hat{N}(k)\right] + f \cdot q_{\mathrm{in}}^{\mathrm{sm}}(k-1) \tag{3}$$

where

$q_{\mathrm{QC}}(k)$ = queue control ramp exit flow,
$f \geq 1$ = factor used to obtain a nonconservative prediction of demand to come in next control period, and
$N_{\max}$ = maximum admissible number of vehicles in ramp queue, which is used as the set point of the queue controller.

Equation 3 should ideally contain the ramp demand $q_{in}(k)$, but, because this demand is unknown at time kT, it must be predicted. An exponentially smoothed value of the flow measurements at the ramp entrance may be used to do this:

$$q_{in}^{sm}(k) = \epsilon q_{in}(k) + (1-\epsilon) q_{in}^{sm}(k-1) \quad (4)$$

where $\epsilon \in [0, 1]$ is the smoothing factor. Thus $q_{in}^{sm}(k-1)$ is used in Equation 3 as a prediction for $q_{in}(k)$.

When ALINEA (or any other local ramp-metering algorithm) is applied to each on-ramp independently without any coordination actions, it may be superseded by queue management actions, in which case a freeway congestion is created that propagates upstream and activates local ramp metering at the next upstream on-ramp as well and so forth. This phenomenon was observed in the field and was discussed in detail in Papamichail et al. (*21*) on the basis of reproducible simulation results with uncoordinated ALINEA application. It was concluded that independent (uncoordinated) application of ALINEA at each ramp (with limited storage space) may ameliorate the traffic conditions (compared with no control) but cannot always eliminate the congestion or minimize its detrimental effects on traffic flow efficiency. This discussion provided a sensible basis for developing the novel and efficient coordination scheme called HERO.

The HERO module in Figure 1 coordinates local ramp-metering actions. Coordination comes into existence by occasional appropriate setting of minimum ramp-queue lengths that should be created and maintained at specific ramps via superseding of local ALINEA actions there. To this end, a suitable queue controller (similar to the one in Equation 3) is used, and a minimum queue control exit flow is calculated for each ramp by the minimum queue control module. The basic philosophy of HERO may be summarized as follows:

1. HERO identifies potentially active mainstream bottlenecks; such bottlenecks where congestion formation is imminent do not need to be preselected but may appear anywhere in the freeway network.
2. To retard or avoid queue management actions at the concerned on-ramp (master) and the resulting mainstream congestion, HERO activates increasingly additional storage space by recruitment of upstream-located slave ramps.
3. The formed cluster of ramps has a much higher total storage capacity than the master ramp by itself; this increased storage space is exploited to avoid or retard the mainstream congestion; the cluster of one master and several slaves is dissolved when the mainstream occupancy at the bottleneck or the master-ramp queue becomes sufficiently low (i.e., when the bottleneck is deactivated).

While HERO is increasingly recruiting slave ramps with corresponding minimum queues, the master ramp's ALINEA continues to operate normally so as to continue to maximize the mainstream throughput at the active bottleneck location.

The final ramp flow specification module is responsible for the choice of the ramp exit flow to be applied at the next control period while the Implementation module calculates the cycle time of the ramp-metering traffic signals that corresponds to the final flow according to the ramp-metering policy used (e.g., one car per green); see Papageorgiou and Papamichail (*29*) for details.

The HERO coordination software is currently being extended with further modules to handle additional issues arising from its practical implementation, such as multiple downstream bottlenecks, waiting-time constraints for vehicles queuing at an on-ramp, and balancing of queues or waiting times on dual-branch on-ramps by appropriate operation of separate signals for each branch.

FIELD APPLICATION

Since the year 2000 Melbourne's freeways have become heavily congested with extended periods of flow breakdown. The Monash Freeway is a six-lane dual carriageway carrying in excess of 160,000 vehicles per day with up to 20% commercial vehicles; it experiences long periods of congestion between 3 to 8 h/day. To address this congestion problem, VicRoads, the responsible road authority for the state of Victoria, Australia, identified the need for

- A systemwide approach to ramp metering that would provide the ability to coordinate on-ramps to balance flows and ramp queues across the network,
- A systemwide approach to resolving freeway bottlenecks by maximizing the available capacity of the freeway under all traffic loading and environmental conditions, and
- A control logic that supports the above goals.

Under this perspective, VicRoads decided to implement the HERO suite of algorithms that aligns with its freeway management objectives.

In early 2008, HERO was operational at six consecutive inbound on-ramps of the Monash Freeway, extending from Jacksons Road to Warrigal Road. This pilot project is part of the Monash–CityLink–West Gate (MCW) upgrade. Significant benefits were demonstrated over the previous ramp-metering system, which was replaced by HERO. The control logic has proven to be robust and transparent to traffic engineers. Transition to HERO has been seamless to motorists and provides significant flexibility and capability to operate the freeway close to optimal conditions. The economic payback period of the pilot project was just 11 days. The successful implementation and evaluation of HERO has led to its rollout during 2009–10 at 63 sites across the entire 75-km route of the MCW upgrade.

Selected field results are presented here to illustrate the basic concepts described in the previous section. Because of the interactions that take place between different modules, results from different days have been selected to demonstrate each concept in the best possible way. Figure 2 displays a graphical representation of the freeway stretch for the pilot project whereby the links of the graph represent freeway stretches with uniform characteristics (i.e., no on- or off-ramps and no major changes in geometry); the nodes of the graph are placed at locations where a major change in road geometry occurs, as well as at off- and on-ramp junctions. Numbered bullets represent the available (but not necessarily HERO-utilized) detector stations.

The ALINEA regulator (Equation 1) is demonstrated during the morning peak of May 21, 2008. During the morning peak, a bottleneck appears downstream of the Warrigal Road on-ramp (ON_WARRIGAL in Figure 2*a*). To address this problem [i.e., maintain the mainstream occupancy at the merge area of ON_WARRIGAL (Detector 7838 in Figure 2*a*) around its set point ($\tilde{o} = 26\%$, dashed line in Figure 3*a*)], ALINEA is acting locally with a gain factor, K_I, set equal to 60 vehicles per hour per percent and is seen in Figure 3 to pursue its objectives quite successfully. The related ramp exit flow ordered by ALINEA is shown in Figure 3*b*. As noted in the previous section, this flow is bounded by the minimum admissible ramp flow, q_{min}, and the ramp's flow capacity under signal control, q_{max}; these parameters were set equal to 400 vehicles per hour and 1,500 veh per hour, respectively (dashed lines in Figure 3*b*).

Queue estimation (Equation 2) is demonstrated for the morning peak of January 25, 2008, at ON_WARRIGAL (Figure 4). If no correction term is used in Equation 2 (i.e., if K_{est} is set to zero), the estimation would be based purely on flow measurement balance and would accumulate all the flow measurement errors (Figure 4*a*).

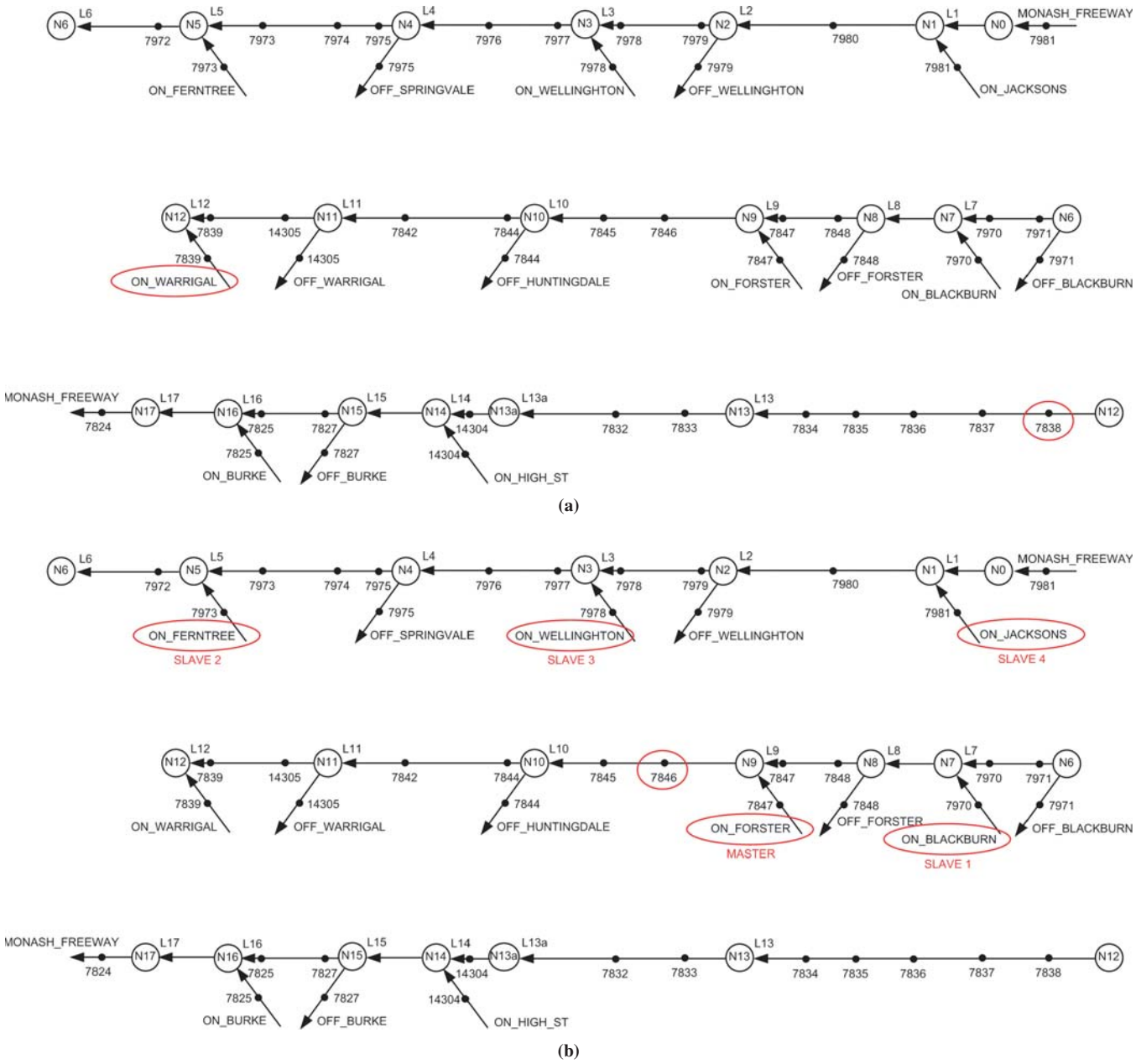


FIGURE 2 Representation of (*a*) pilot project freeway stretch with ALINEA applied at ON_WARRIGAL and (*b*) pilot project freeway stretch with HERO coordination.

In contrast, if K_{est} is set to 0.1 (as in the actual application), then occupancy measurements from the middle of the ramp (Figure 4*b*) are also used and the estimation becomes more accurate. With a bumper-to-bumper capacity of the ramp, N^{BB}, equal to 56 vehicles, an electrical detector length, ϵ_{el}, equal to 1 m and a physical vehicle length, L_{ph}, equal to 5 m, the resulting ramp queue estimates are displayed in Figure 4*d*. As both Figures 4*a* (dashed line) and 4*d* (upper dashed line) show, some produced estimates were eventually bounded; the lower bound is of course zero while the upper bound, N^{cap}, was set equal to 46 vehicles. Figure 4*d* shows that the queue was maintained around the value of 28 vehicles (lower dashed line), which was used in this case as a set point, N_{max}, for the queue controller (Equation 3). This set point corresponds to an almost fully occupied ramp, as can be verified from the peaks observed in the occupancy measurements at the entrance of the ramp whenever the queue is around 28 vehicles (Figure 4*c*).

The queue override policy is demonstrated for the morning peak of 3 March 2008 at ON_WARRIGAL (Figure 5). The threshold, o_{th}, was set to 80% (dashed line in Figure 5*a*). Whenever the occupancy measurements at the entrance of the ramp exceeded that threshold, the queue override ramp exit flow, $q_{QO}(k)$, was set to $q_{over} = 1{,}200$ vehicles per hour (Figure 5*b*).

The queue control (Equation 3) is demonstrated through its activation during the morning peak of 25 January 2008 at ON_WARRIGAL (Figure 6). The corresponding ramp-queue length estimation is shown in Figure 6*a*. The set point, N_{max}, used by the queue controller is set to 28 vehicles, while the *f* factor is set to 1.1. One can see that the ramp-queue length is maintained, when necessary, around its set point with proper action of the queue controller. The corresponding ramp exit flow ordered by the queue control is given in Figure 6*b*. Negative flows may result from the queue controller (Equation 3), as

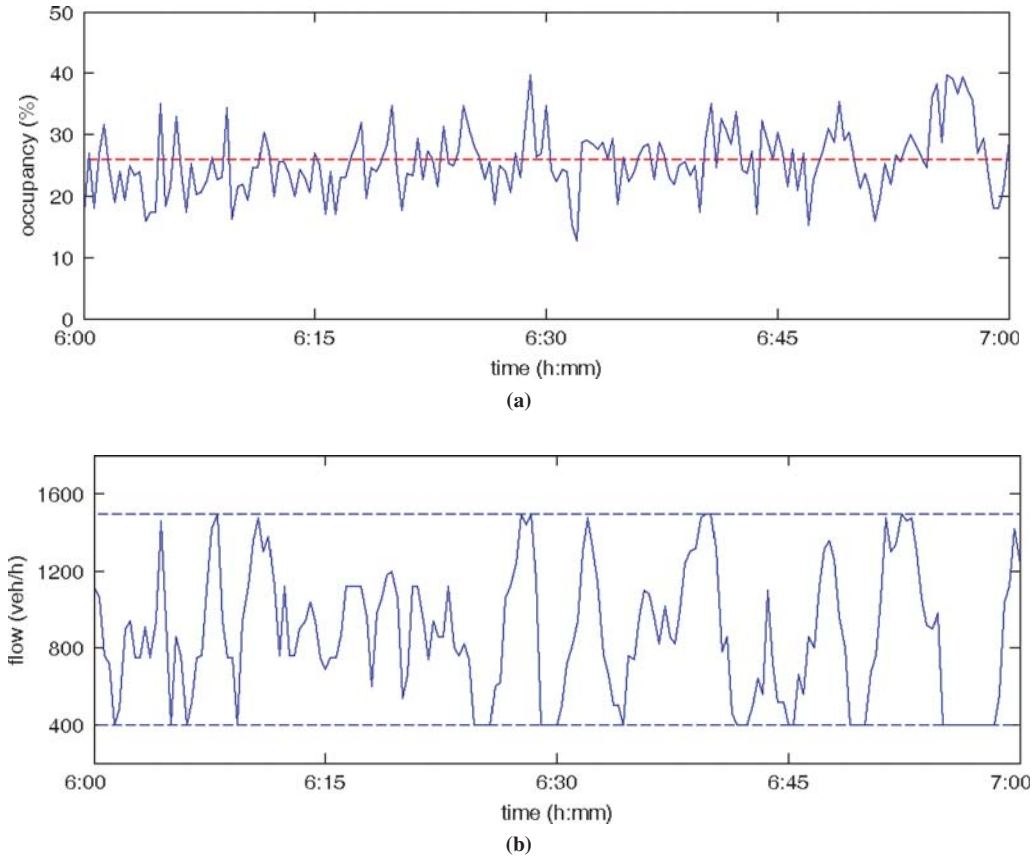


FIGURE 3 ALINEA application at ON_WARRIGAL: (*a*) occupancy measurement at Detector 7838 and (*b*) flow ordered by ALINEA.

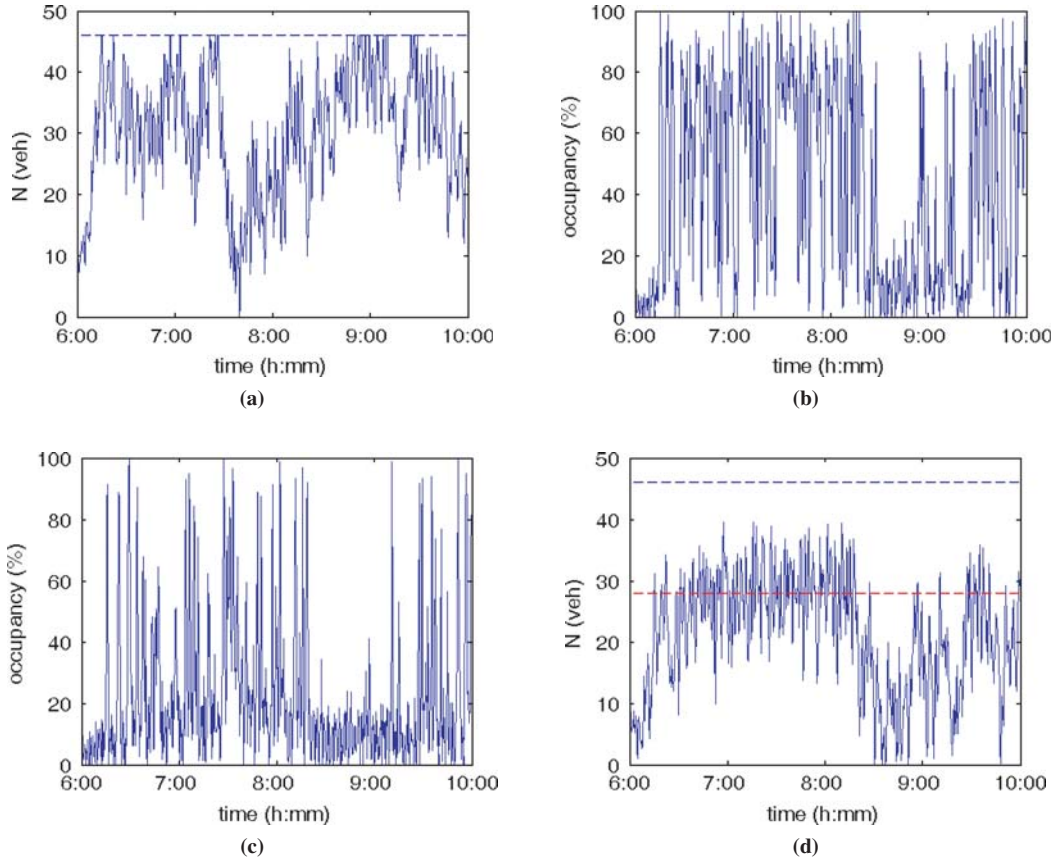


FIGURE 4 Queue estimation at ON_WARRIGAL: (*a*) estimation based on flow balance, (*b*) occupancy at midpoint, (*c*) occupancy at entrance, and (*d*) estimation based on Kalman filter.

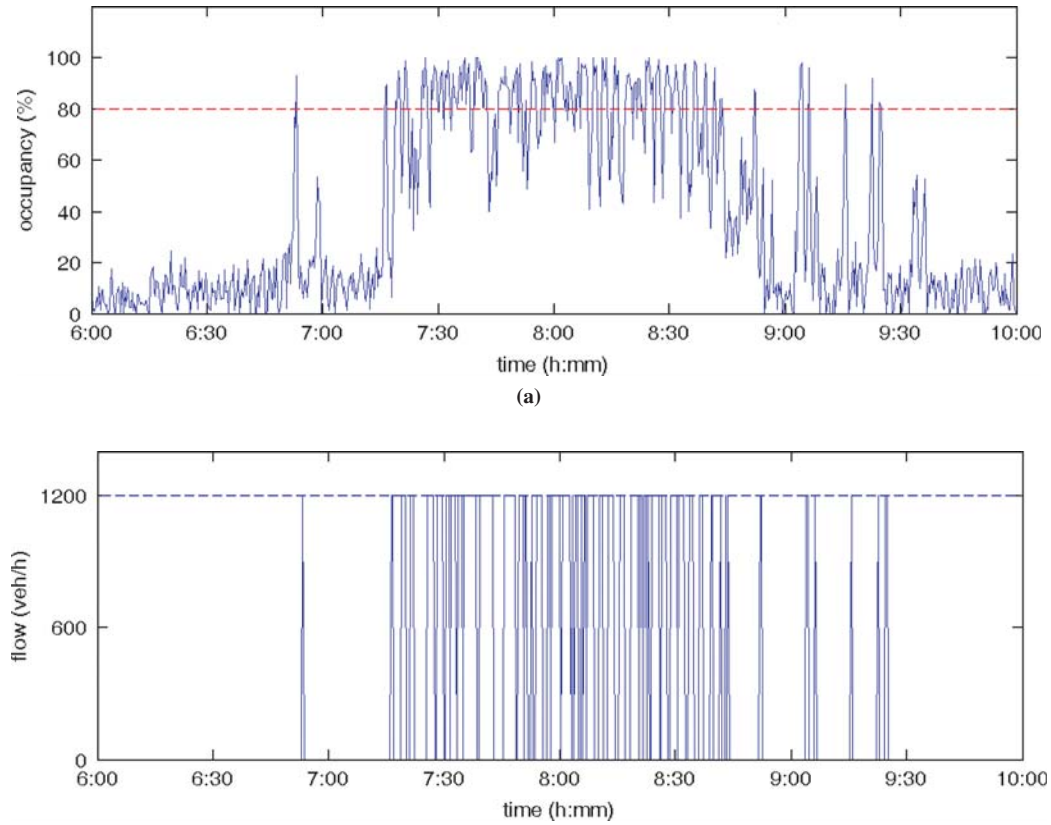


FIGURE 5 Queue override at ON_WARRIGAL: (*a*) occupancy measurement at ramp entrance and (*b*) ramp exit flow ordered by queue override.

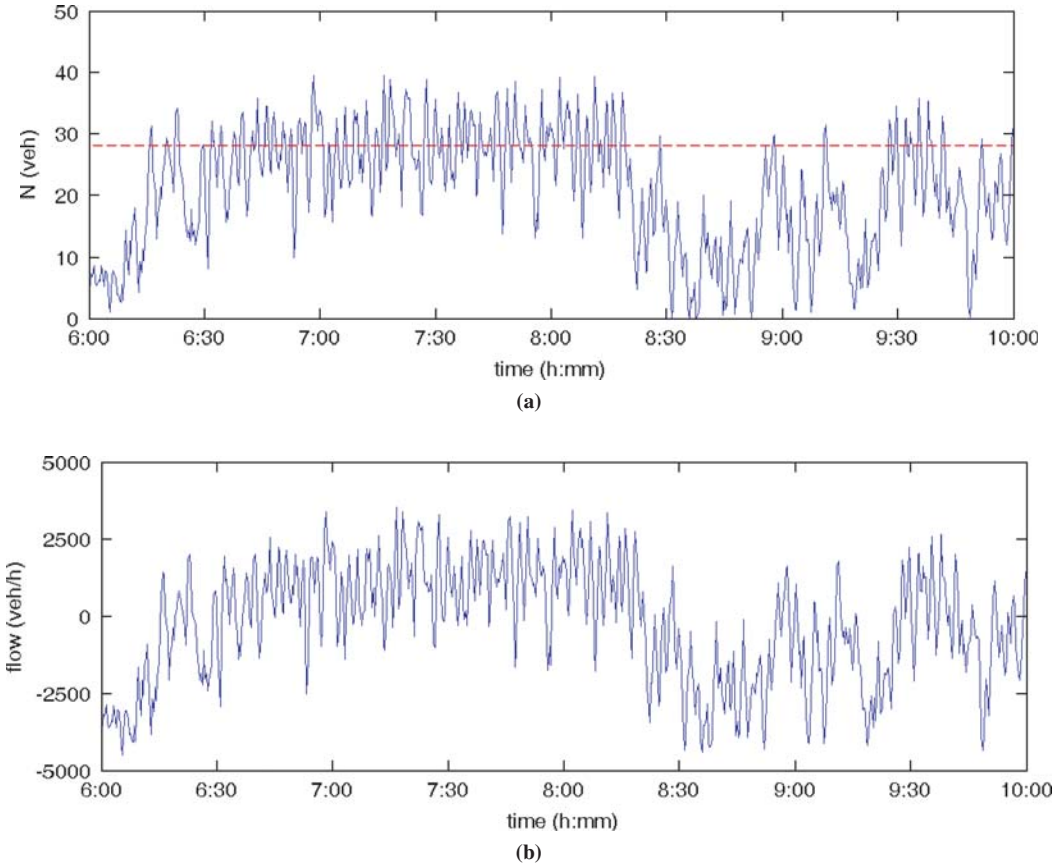


FIGURE 6 Queue control at ON_WARRIGAL: (*a*) estimation of ramp queue length and (*b*) ramp exit flow order by queue control.

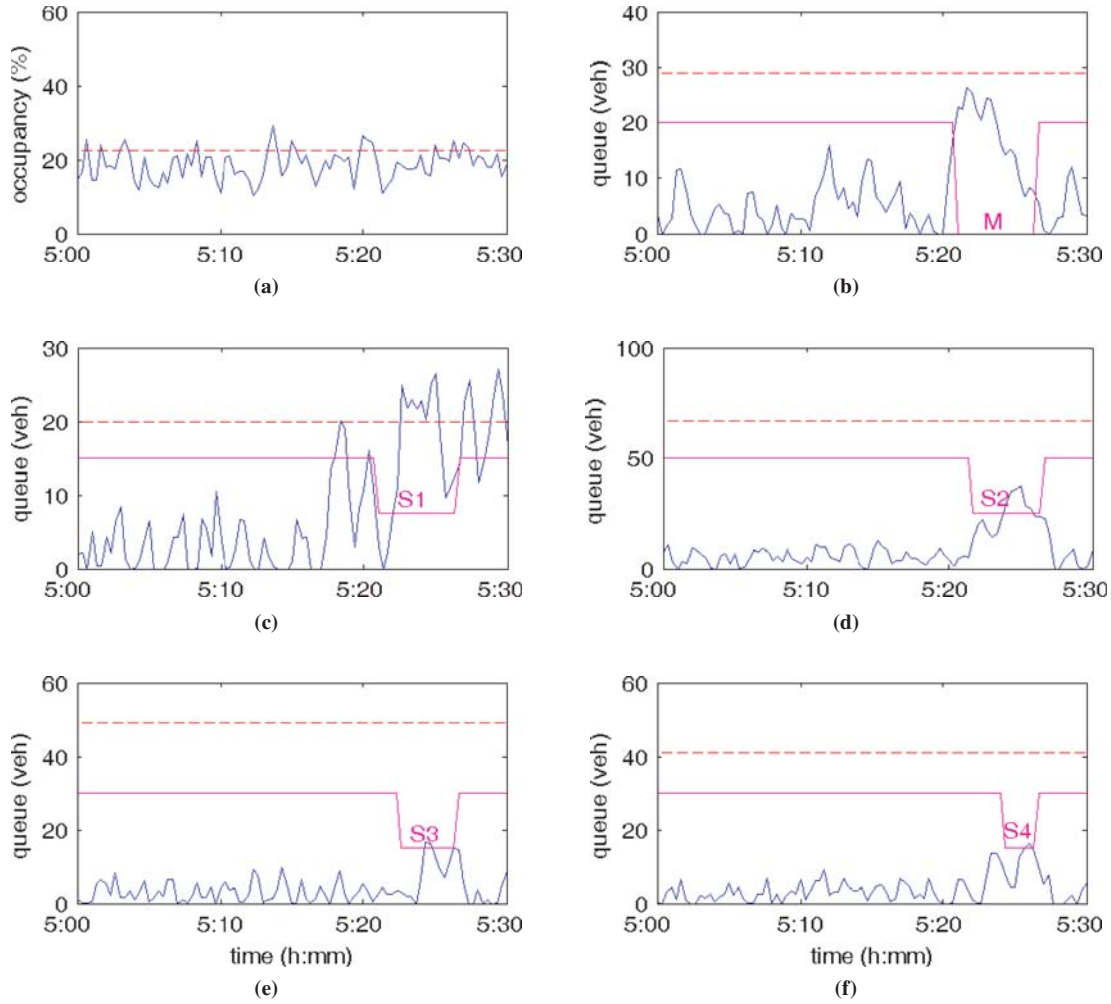


FIGURE 7 HERO coordination actions: (*a*) occupancy ON_FORSTER at Detector 7846, (*b*) queue ON_FORSTER, (*c*) queue ON_BLACKBURN, (*d*) queue ON_FERNTREE, (*e*) queue ON_WELLINGTON, and (*f*) queue ON_JACKSONS.

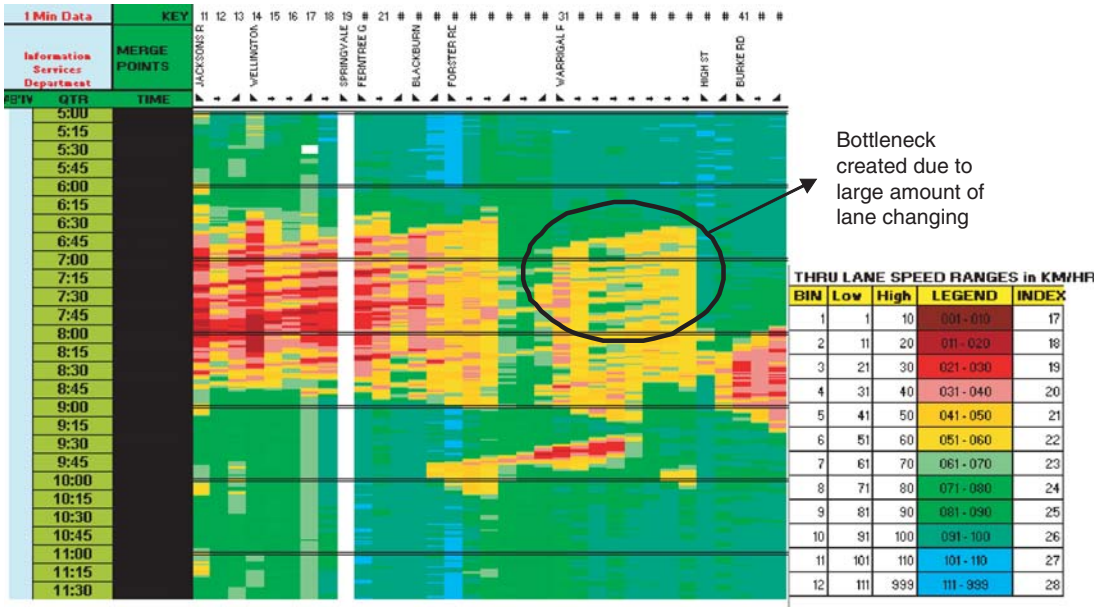
in parts of Figure 6*b,* when the number of vehicles in the ramp is low, but this has obviously no implications for the final ramp-metering outcome.

HERO's coordination actions are demonstrated through its activation during the afternoon peak of 28 March 2008. During the afternoon peak, a bottleneck (circled in Figure 2*b*) is imminent just downstream of the Forster Road on-ramp (ON_FORSTER in Figure 2*b*). To address the problem [i.e., maintain the mainstream occupancy at the merge area of ON_FORSTER (the area of Detector 7846) around its set point (dashed line in Figure 7*a*)], ALINEA is acting locally and short queues are forming as shown in Figure 7*b.* At 5:21 p.m., the queue on ON_FORSTER exceeds a prespecified threshold value and HERO is activated. ON_FORSTER becomes a master (MASTER in Figure 2*b* and M in Figure 7*b*) and recruits the first upstream on-ramp (i.e., ON_BLACKBURN) as its first slave ramp (SLAVE 1 in Figure 2*b* and S1 in Figure 7*c*). The queue that formed at ON_BLACKBURN just before 5:20 p.m. (and dissolved shortly after) was due to local control actions of ALINEA at that ramp. After the recruitment of the ramp as a slave, ALINEA at ON_BLACKBURN is superseded and a ramp queue is deliberately created and maintained as a result of coordination. The maximum queue allowed on each on-ramp is represented by a dashed line in Figures 7*b* through 7*f* and is the set point used by the queue controllers of the respective on-ramps.

The bottleneck at the merge area of ON_FORSTER persists and, as a result, more upstream on-ramps are gradually recruited by HERO as slaves (S2, S3, and S4 in Figure 7). The small queue values appearing at ON_FERNTREE, ON_WELLINGTON, and ON_JACKSONS before their recruitment as slave ramps reflect the respective numbers of moving vehicles on these on-ramps. The formed cluster of ramps (M, S1, S2, S3, and S4) is dissolved at 5:27 p.m. as the queue on the master ramp becomes sufficiently low. Thus, the imminent bottleneck activation downstream of the ON_FORSTER ramp could be avoided by activation of ramp storage space available at five on-ramps, thanks to the coordinating HERO actions (and the local ALINEA actions).

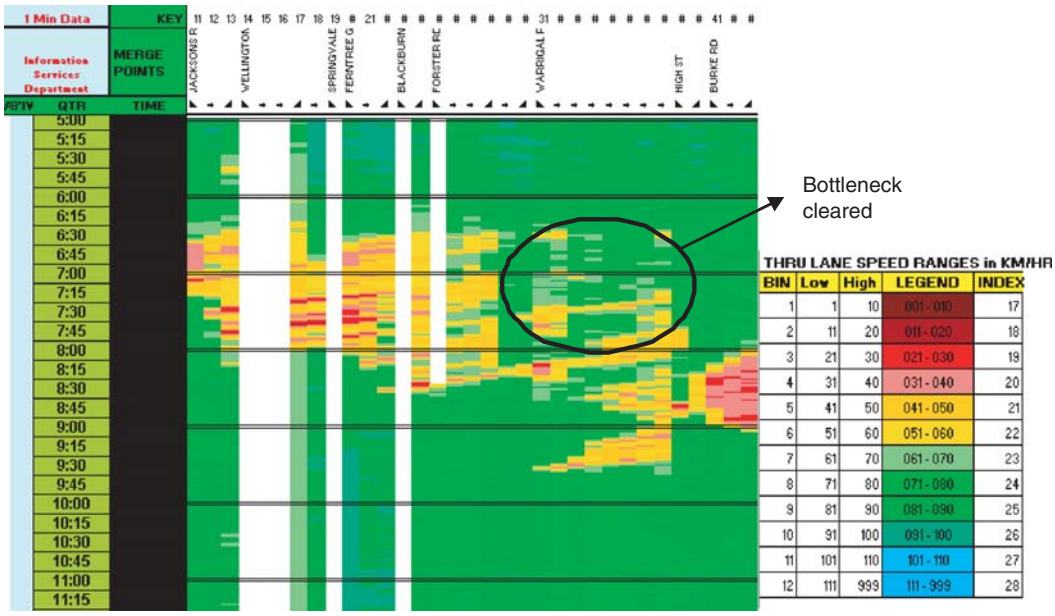
GLOBAL EVALUATION RESULTS

An evaluation of HERO's field performance was undertaken by VicRoads for the pilot project mentioned earlier. Before-and-after speed contour plots based on morning operation in 2007 and 2008 are shown in Figures 8*a* and 8*b.* The before case involves a



BIN	Low	High	LEGEND	INDEX
1	1	10	001 - 010	17
2	11	20	011 - 020	18
3	21	30	021 - 030	19
4	31	40	031 - 040	20
5	41	50	041 - 050	21
6	51	60	051 - 060	22
7	61	70	061 - 070	23
8	71	80	071 - 080	24
9	81	90	081 - 090	25
10	91	100	091 - 100	26
11	101	110	101 - 110	27
12	111	999	111 - 999	28

(a)



BIN	Low	High	LEGEND	INDEX
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4	31	40	031 - 040	20
5	41	50	041 - 050	21
6	51	60	051 - 060	22
7	61	70	061 - 070	23
8	71	80	071 - 080	24
9	81	90	081 - 090	25
10	91	100	091 - 100	26
11	101	110	101 - 110	27
12	111	999	111 - 999	28

(b)

FIGURE 8 Typical morning speed contour plot for (*a*) previous metering algorithm and (*b*) HERO.

ramp-metering system that was in use before HERO. HERO sensibly reduced the space–time extent of the freeway traffic flow breakdown and provided significant improvements in throughput and travel speed compared with the previously used ramp-metering system.

The morning peak evaluation (6:00 to 9:00 a.m.) revealed a 4.7% increase in average flow on the freeway between Warrigal Road and High Street (on top of the previous system) and a 35% increase in average speed; while the afternoon peak evaluation (5:30 to 6:30 p.m.) showed an 8.4% increase in average flow and a 58.6% increase in average speed. Comparisons involving the Austroads National Performance Indicators (ANPI) (*30*) are displayed in Figure 9 for the morning peak. Very significant improvements are observed in

- Productivity (a specific ANPI reflecting a combination of high speed and high volume on the freeway),
- Mean speed deviation (due to congestion) from the posted (maximum admissible) speed, and
- Reliability (a specific ANPI reflecting travel time differences from day to day).

For more details on global evaluation results, the reader is referred to a recent paper by Vong and Gaffney (*31*).

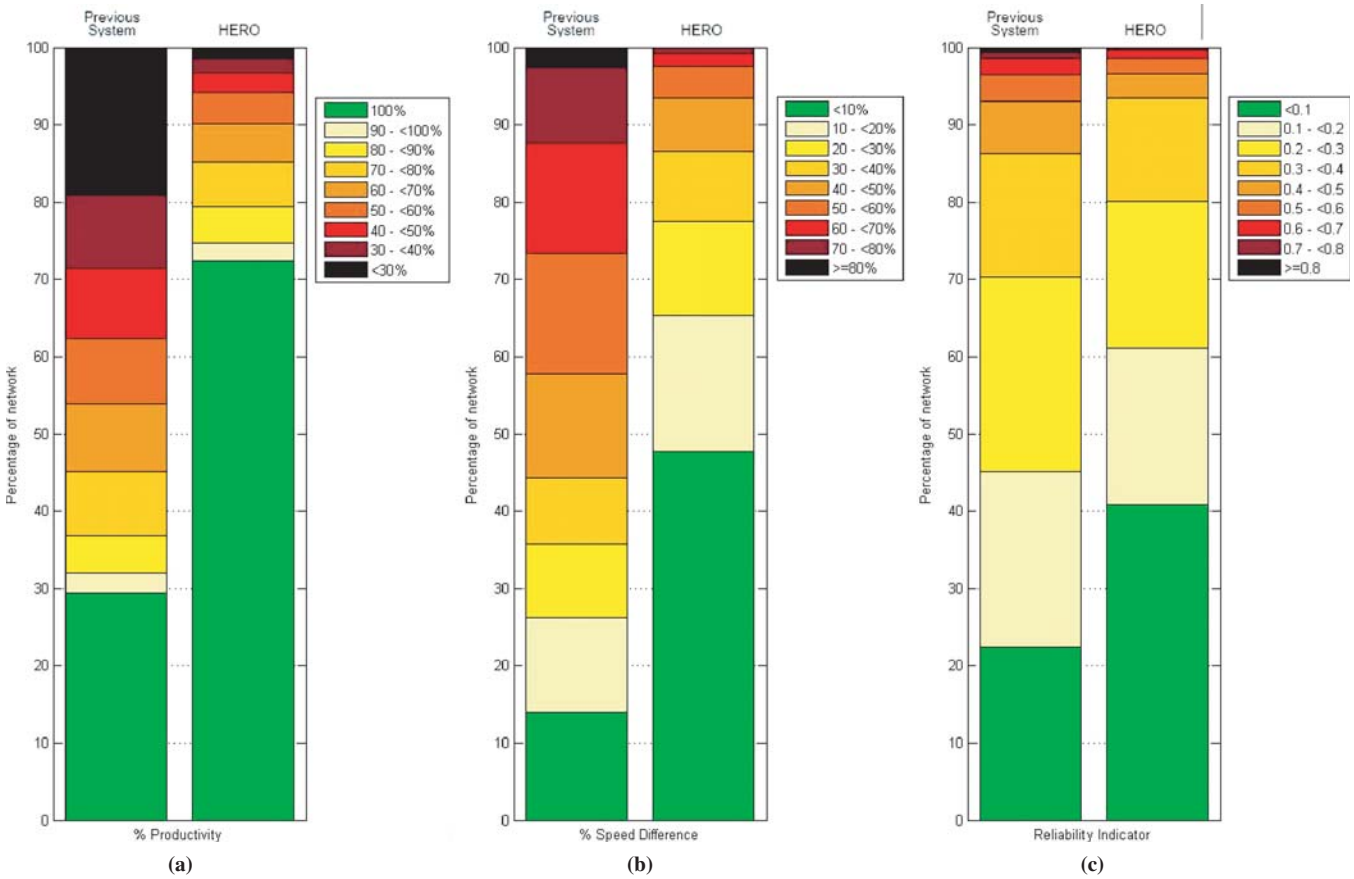

FIGURE 9 Austroads national performance indicators, previous system versus HERO, for (*a*) percent productivity, (*b*) percent speed difference, and (*c*) reliability indicator.

CONCLUSIONS

HERO, a new heuristic traffic-responsive feedback control strategy that coordinates local ramp-metering actions for freeway networks, has been presented in this paper. The proposed coordination scheme is simple and reactive (i.e., based on readily available real-time measurements without the need for real-time model calculations or external disturbance prediction). HERO employs an extended version of the feedback regulator ALINEA at a local level. The new strategy overcomes the problem of uncertain freeway capacity by targeting the critical occupancy for maximum throughput. HERO outperforms uncoordinated local ramp metering and approaches the efficiency of sophisticated optimal control schemes.

HERO has been implemented by VicRoads during a pilot project at six consecutive inbound on-ramps on the Monash Freeway in Melbourne, Australia, and was evaluated against the previously used ramp-metering system. The economic payback period of the pilot project was just 11 days. The results show a significant increase of traffic throughput and a reduction of travel times. The successful implementation and evaluation of HERO has led to its rollout during 2009–10 at 63 sites across the entire 75-km route of the MCW upgrade.

HERO was also field implemented in a 20-km stretch of the inbound A6 freeway in the south of Paris, France, in 2006, albeit in a simplified form due to lack of real-time on-ramp data in the control center; nevertheless, results indicated a clear improvement over the uncoordinated ALINEA case. In addition, HERO has been adopted for field implementation at all urban on-ramps (some 40) of the ring-road freeway A10 around Amsterdam, the Netherlands, by the responsible road authority (Rijkswaterstaat), and the related implementation is in the planning phase. Finally, several other authorities in North America and Australia have expressed their strong interest in adopting HERO for their respective freeway networks.

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